Multiple Linear Regression

1. Definition

Multiple Linear Regression is a supervised learning algorithm used to predict a continuous dependent variable using two or more independent variables.

It establishes a linear relationship between input variables and output variable.

2. Mathematical Formula

Y=b0 + b1X1 + b2X2 + b3X3+...+ bnXn

Where:

- YYY = Dependent variable (Output)
- b0= Intercept
- b1,b2,b3 = Coefficients
- X1, X2, X3 = Independent variables

3. Case Study: Student Marks Prediction

Problem Statement

Predict **Student Final Marks** based on:

- Study Hours
- Attendance (%)
- Internal Marks

This is a **Multiple Linear Regression** problem because:

1) There are **multiple independent variables**
2) One dependent variable (Final Marks)

Predict **Final Marks (dependent)** based on:

- Study Hours (X1)
- Attendance (X2)
- Internal Marks (X3)

Sample Data:

Study Hours	Attendance (%)	Internal Marks	Final Marks
2	80	15	45
3	85	18	50
5	75	14	55
6	90	22	70
8	95	25	85
4	70	12	48
7	88	20	78

Python Code

```
# Import libraries
import pandas as pd
from sklearn.linear_model import LinearRegression

# Create dataset
data = {
    'StudyHours': [2,3,5,6,8,4,7],
    'Attendance': [80,85,75,90,95,70,88],
    'InternalMarks': [15,18,14,22,25,12,20],
    'FinalMarks': [45,50,55,70,85,48,78]
}

df = pd.DataFrame(data)

# Independent variables (X)
X = df[['StudyHours', 'Attendance', 'InternalMarks']]

# Dependent variable (y)
y = df['FinalMarks']

# Create model
model = LinearRegression()

# Train model
model.fit(X, y)

# Print coefficients
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)

# Predict values
predictions = model.predict(X)
print("Predicted Marks:", predictions)
```

Sample Output (Approximate)

```
Intercept: -5.20
Coefficients: [3.10, 0.45, 1.80]

Predicted Marks:
[44.2, 50.1, 54.8, 69.9, 85.4, 47.5, 77.8]
```

Output Explanation

Regression Equation

Multiple Linear Regression formula: **Y=b0 + b1 X1 + b2 X2 + b3 X3**

From output:

Final Marks = −5.20 + (3.10×StudyHours) + (0.45×Attendance) + (1.80×InternalMarks)

Meaning of Coefficients

◆ **Study Hours (b1=3.10)**

If study hours increase by 1 hour, final marks increase by **3.10 marks**, keeping other variables constant.

◆ **Attendance (b2=0.45)**

If attendance increases by 1%, marks increase by **0.45 marks**.

◆ **Internal Marks (b3= 1.80)**

If internal marks increase by 1 mark, final marks increase by **1.80 marks**.

Intercept (b0=-5.20) -

If all independent variables are zero, predicted marks = -5.20
(Practically not meaningful, but mathematically required)

Predicted vs Actual Values

Actual Final Marks	Final MarksPredicted by model
45	44.2
50	50.1
55	54.8
70	69.9

Actual Final Marks	Final MarksPredicted by model
85	85.4

Predicted values are very close to actual values.

This means the model fits well.

Advantages

- Gives relationship between variables
- Helps in prediction
- Easy to implement
- Interpretable model

Conclusion

Multiple Linear Regression helps in predicting a continuous value by considering multiple factors. It provides coefficients that show the impact of each independent variable on the dependent variable.

This is Multiple Linear Regression

Because:

- More than one independent variable
- Linear relationship
- Continuous output variable

✓ Important Technical Viva Questions

1. Why is this regression called multiple?
2. What is the regression equation?
3. What does coefficient represent?
4. What happens if we remove one feature?
5. What assumptions does linear regression follow?

★ 10-Mark Answer Structure for SPPU Exam

1. Define Multiple Linear Regression
2. Write formula
3. Give dataset example
4. Show equation
5. Explain coefficients
6. Interpret output